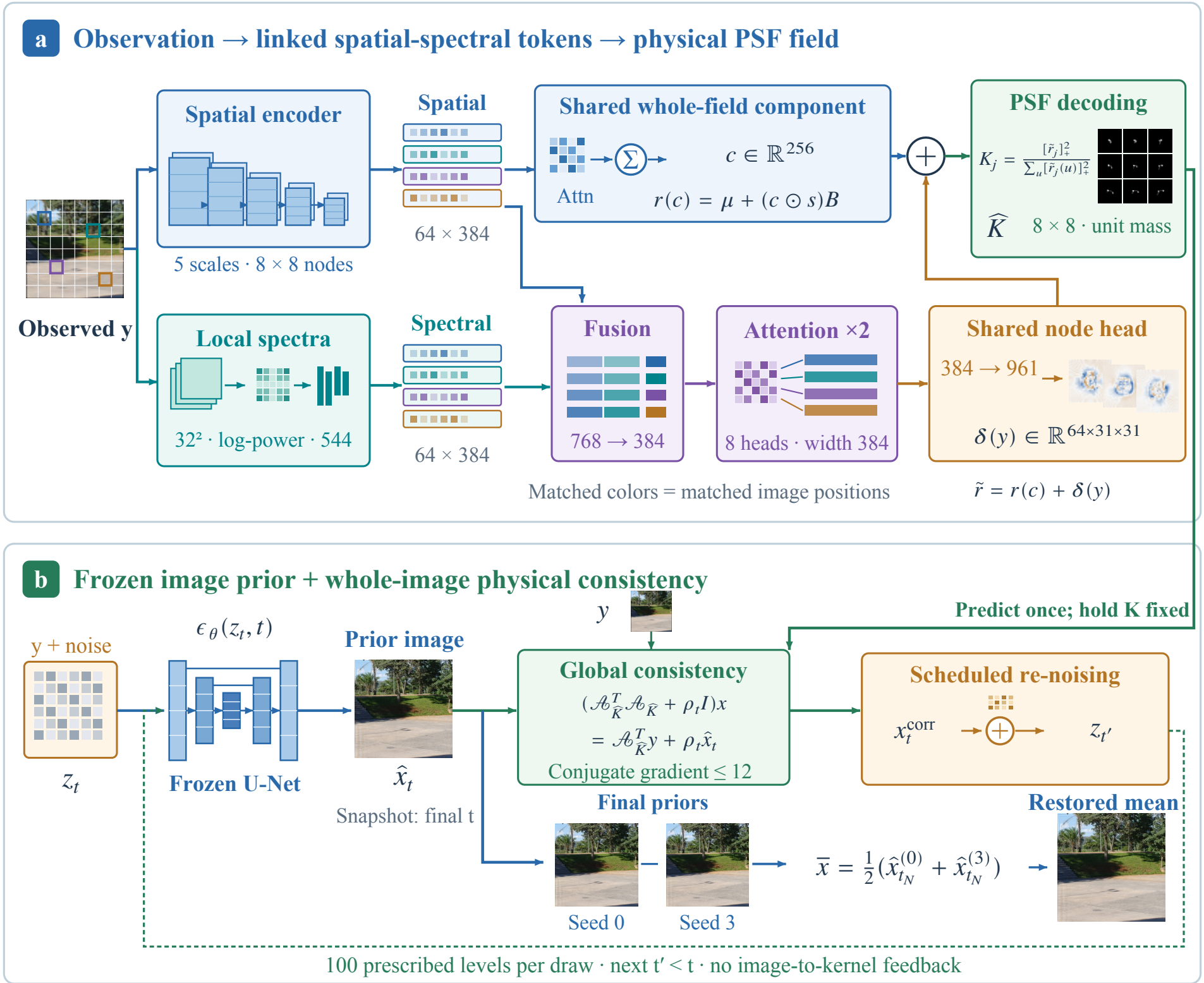

a
Observation → linked spatial-spectral tokens → physical PSF field
Observed y
Spatial encoder
5 scales · 8 × 8 nodes
Spatial
64 × 384
Shared whole-field component
Attn
$c \in \mathbb{R}^{256}$
$r(c) = \mu + (c \odot s)B$
PSF decoding
$K_j = \frac{[\tilde{r}_j]_+^2}{\sum_u [\tilde{r}_j(u)]_+^2}$
$\widehat{K}$
8 × 8 · unit mass
Local spectra
32² · log-power · 544
Spectral
64 × 384
Fusion
768 → 384
Attention ×2
8 heads · width 384
Shared node head
384 → 961
$\delta(y) \in \mathbb{R}^{64\times31\times31}$
Matched colors = matched image positions
$\tilde{r} = r(c) + \delta(y)$
b
Frozen image prior + whole-image physical consistency
y + noise
z_t
$\epsilon_\theta(z_t, t)$
Frozen U-Net
Prior image
$\hat{x}_t$
Snapshot: final t
y
Global consistency
$(\mathcal{A}_{\widehat{K}}^T \mathcal{A}_{\widehat{K}} + \rho_t I)x$
$= \mathcal{A}_{\widehat{K}}^T y + \rho_t \hat{x}_t$
Conjugate gradient ≤ 12
Predict once; hold K fixed
Scheduled re-noising
x_t^{corr}
$z_{t'}$
Final priors
Seed 0
Seed 3
$\bar{x} = \frac{1}{2}(\hat{x}_{t_N}^{(0)} + \hat{x}_{t_N}^{(3)})$
Restored mean
100 prescribed levels per draw · next t′ < t · no image-to-kernel feedback